• Science & Technology

61. With reference to the use of nano-technology in health sector, which of the following statements is/are correct?

1. Targeted drug delivery is made possible by nanotechnology.
2. Nanotechnology can largely contribute to gene therapy.

Select the correct answer using the code given below:

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (c) Nanoparticle can be used in targeted drug delivery at the site of disease to improve the uptake of poorly soluble drugs.
In gene therapy using the non-viral system, the main issues are relatively transient gene expression and lower efficiency than viral vectors. Research efforts have focused on understating the barriers in gene delivery so that non-viral systems can be developed that are as effective as viral systems in gene transfixion.

62. With reference to `Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields.
2. NFC is designed for use by devices which can be at a distance of even a metre from each other.
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (c) Near Field Communication (NFC) is the set of protocols that enable the electronic devices to establish radio communication with each other by touching the devices together or bringing them into proximity to a distance of typically 10 cm or less. There can be secure communications by applying an encryption algorithm as it is done for credit cards and if it fits the criteria for being considered a personal area network. Hence option (c) is correct.

63. The term `Goldilocks Zone' is often seen in the news in the context of–

(a) the limits of habitable zone above the surface of the Earth
(b) regions inside the Earth where shale gas is available
(c) search for the Earth-like planets in outer space
(d) search for meteorites containing precious metals

Ans : (c) Goldilocks Zone is an area in space where a planet is so far away from its stars that the surface of that plant is neither too hot nor too cold. In the news, we often find references to the discovery of Earth- like planets in outer space.

64. The Genetic Engineering Appraisal Committee is constituted under the

(a) Food Safety and Standards Act, 2006
(b) Geographical Indications of Goods (Registration and Protection) Act, 1999
(c) Environment (Protection) Act, 1986
(d) Wildlife (Protection) Act, 1972

Ans : (c) The genetic engineering appraisal committee (CEAC) is the Apex body Constituted in the Ministry of Environment and Forests. The Genetic Engineering Appraisal Committee has been constituted under the Environment (Protection) Act, 1986 and the ministry of Environment and forest.

65. In which of the following activities are Indian Remote Sensing (IRS) satellites used?

1. Assessment of crop productivity
2. Locating groundwater resources
3. Mineral exploration
4. Telecommunications
5. Traffic studies

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 4 and 5 only
(c) 1 and 2 only (d) 1, 2, 3, 4 and 5

Ans : (a) The Indian Space Programme has two major operating systems- the Indian National Satellite (INSAT), which is a geostationary satellite is used for telecommunication, television broadcasting and meteorological services and the Indian Remote sensing satellite (IRS) for monitoring and management of natural resources (e.g. locating groundwater resources, mineral exploration etc.) and disaster management support.

ENVIRONMENT & ECOLOGY

66. With reference to 'dugong', a mammal found in India, which of the following statements is/are correct?

1. It is herbivorous marine animal.
2. It is found along the entire coast of India.
3. It is given legal protection under Schedule I of the Wildlife (Protection) Act, 1972

Select the correct answer using the code given below–

(a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only

Ans : (c) Feeding on Seagrass, dugongs are found in Seagrass beds, sheltered waters, lagoons and bays. In India, it is found off the Gujarat coast (Gulf of Kutch). Tamil Nadu Coast (Gulf of Mannar), Palk Bay and Andaman and Nicobar Islands. Dugongs had been brought under schedule I of the Wildlife Protection Act.

67. Which one of the following is the national aquatic animal of India?

(a) Saltwater crocodile (b) Olive ridley turtle
(c) Gangetic dolphin (d) Gharial

Ans : (c) On 5th October, 2009, the Ganga dolphin has been declared a national aquatic animal by the central government. They are locally known as Sush, because of the noise it makes while breathing. Ganga Dolphin is a critically endangered species in India and therefore, has been included in schedule I for the wildlife (protection) Act 1972.

68. Which of the following National Parks is unique in being a swamp with floating vegetation that supports a rich biodiversity?

(a) Bhitarkanika National Park
(b) Keibul Lamjao National Park
(c) Keoladeo Ghana National Park
(d) Sultanpur National Park

Ans : (b) Keibul Lamjao National Park is the world's only floating national park and the last natural habitat of Manipur Brow antlered deer-Sangai (Cervus eldi eldi). The park was established in the year 1977. It is spread over 40 sq.km of area in the Vishnupur district of Manipur.

69. With reference to an organization known as 'BirdLife International', which of the following statements is/are correct?

1. It is a Global Partnership of Conservation Organizations.
2. The concept of `biodiversity hotspots' originated from this organization.
3. It identifies the sites known/referred to as `Important Bird and Biodiversity Areas'.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (c) Important Bird and Biodiversity Area (IBA) programme aims to identify, monitor & protect a global network of IBAs for the conservation of the worlds birds and other wildlife. The concept of 'Biodiversity hotspots' did not originate from this organisation. It was Coined by Norman Meyers in 1988. Each Biodiversity hotspot represents a remarkable universe of extraordinary floral and faunal endemicity struggling to service in the rapidly Shrinking ecosystem. Hence option (c) is correct.

70. 'BioCarbon Fund Initiative for Sustainable Forest Landscapes' is managed by the–

(a) Asian Development Bank
(b) International Monetary Fund
(c) United Nations Environment Programme
(d) World Bank

Ans : (d) The BioCarbon Fund Initiative for sustainable forest landscapes (ISFL) is a multilateral fund, Supported by donor governments and managed by the World Bank. It seeks to promote reduced Green House Gas emissions from the land sector from deforestation and forest degradation in developing countries (REDD+) and from sustainable agriculture, as well as smarter land-use planning, policies and practices.

71. Which of the following statements is/are correct regarding National Innovation Foundation-India (NIF)?

1. NIF is an autonomous body of the Department of Science and Technology under the Central Government.
2. NIF is an initiative to strengthen the highly advanced scientific research in India's premier scientific institutions in collaboration with highly advanced foreign scientific institutions.

Select the correct answer using the code given below :

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (a) National Innovation Foundation India is an autonomous body of the Department of Science and Technology, Government of India, based on the Honey Bee Network philosophy. Which provides institutional support to grassroots innovators and outstanding traditional knowledge holders from the unorganized sector of the society. So, statement 1 is correct and statement 2 is incorrect.

72. Which one of the following issues the `Global Economic Prospects' report periodically?

(a) The Asian Development Bank
(b) The European Bank for Reconstruction and Development
(c) The US Federal Reserve Bank
(d) The World Bank

Ans : (d) World Bank publishes the 'Global Economic Prospects' report periodically.

73. With reference to the International Union for Conservation of Nature and Natural Resources (IUCN) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), which of the following statements is/are correct?

1. IUCN is an organ of the United Nations and CITES is an international agreement between governments.
2. IUCN runs thousands of field projects around the world to better manage natural environments.
3. CITES is legally binding on the States that have joined it, but this Convention does not take the place of national laws.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (b) IUCN is a voluntary organisation, not an agency of the United Nations. This eliminates the other 3 options.
It was founded in 1948 as the world's first global environmental organisation. IUCN supports scientific research, manages field projects all over the world and brings governments, NGOs the UN and companies together to develop policy laws and best practise.

74. With reference to `fly ash' produced by the power plants using coal as fuel, which of the following statements is/are correct?

1. Fly ash can be used in the production of bricks for building construction.
2. Fly ash can be used as a replacement for some of the Portland cement contents of concrete.
3. Fly ash is made up of silicon dioxide and calcium oxide only, and does not contain any toxic elements.

Select the correct answer using the code given below :

(a) 1 and 2 (b) 2 only
(c) 1 and 3 (d) 3 only